

Tanuj Kumar

Ph.D. Candidate

Department of Electrical and Computer Engineering

University of Wisconsin-Madison

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[Google Scholar](#)

Education

University of Wisconsin-Madison

Fall 2021 - *

Ph.D., Electrical & Computer Engineering

M.S., Electrical & Computer Engineering (2023)

4.0/4.0

Advisor: Dr. Mikhail Kats

Birla Institute of Technology & Science (BITS), Pilani, India

2015 - 2020

Integrated Dual Degree program

7.96/10

Bachelor of Engineering (Honours) in Electrical and Electronics

Master of Science (Honours) in Chemistry

Awards

1. David G. Walsh Travel Award, McPherson Eye Research Institute, UW-Madison (2024)
2. Graduate Student Service Award, Department of Electrical and Computer Engineering, UW-Madison (2024)
3. Marie Christine Kohler Fellowship to explore the intersection of science and art at the Wisconsin Institute for Discovery, UW-Madison (2023)
4. Student Research Grants Competition (SRGC) Research Travel Award, UW-Madison (2022)
5. ECE Chancellors Opportunity Fellowship (COF) award, UW-Madison (2021)

Journal Publications

1. S. Yin, J-W Cho, D. Feng, H. Mei, **T. Kumar**, C. Wan, Y. Jin, M. Kim, M.A. Kats, *Preventing overfitting in infrared ellipsometry using temperature dependence: fused silica as a case study*, Apr 2024. [\[Preprint\]](#)
2. D. Feng*, **T. Kumar***, S. Yin, M. Mah, P. Lin, M. Fortman, G.R. Jaffe, C. Wan, H. Mei, Y. Xiao, R. Synowicki, R.J. Warzoha, V.W. Brar, J.J. Talghader, M.A. Kats, *Self-referencing photothermal common-path interferometry to measure absorption of Si_3N_4 membranes for laser-light sails*, Apr 2024. [\[Preprint\]](#)
3. **T. Kumar**, M.A. Kats, *ChatGPT-4 with Code Interpreter can be used to solve introductory college-level vector calculus and electromagnetism problems*, American Journal of Physics, Sep 2023. [\[Article\]](#)
4. **T. Kumar**, R.E. Owyung, S. Sonkusale, *Rapid Cleanroom-Free Fabrication of Thread Based Transistors Using Three-Dimensional Stencil-Based Patterning*, Flexible and Printed Electronics, Feb 2021. [\[Article\]](#)

Patents

1. S. Sonkusale, R.E. Owyung, **T. Kumar**, *Fabrication of Thread Based Transistors*, provisional patent application filed Jan 2021.

Conference Presentations

1. **T. Kumar**, D. Feng, S. Yin, P. Lin, M. Mah, M. Fortman, C. Wan, C. Fang, K. Schnittker, J.B. Andrews, R. Warzoha, V. W. Brar, J. J. Talghader, M. A. Kats, *Interferometric Measurements of Temperature-Dependent Optical Absorption in Low-Loss SiN Membranes for Laser Sails*, Materials Research Society (MRS) Fall Meeting, Boston (Nov 2024, accepted).
2. J.-W. Cho, J.L. King, D. Quach, M. Hafermann, K. Paz, H. Mei, S. Yin, **T. Kumar**, J. Andrade, C. Hessel, C. Ronning, D. Woolf, M. A. Kats, *Engineering of vanadium dioxide for reconfigurable optics by combining metal doping and defect engineering*, Advanced Photonics Congress, Montreal (Jul 2024).
3. J. L. King, J.-W. Cho, D. Quach, M. Hafermann, K. Paz, H. Mei, S. Yin, **T. Kumar**, J. Andrade, C. Hessel, C. Ronning, D. Woolf, M. A. Kats, *Impact of lattice strain on vanadium dioxide transition temperature, induced via defects and doping*, Advanced Photonics Congress, Montreal (Jul 2024).
4. **T. Kumar**, D. Feng, S. Yin, P. Lin, M. Mah, M. Fortman, G. R. Jaffe, C. Wan, C. Fang, R. Warzoha, V. W. Brar, J. J. Talghader, M. A. Kats, *Optical Loss Measurements of Silicon-Nitride Membranes using Photothermal Common-Path Interferometry*, Materials Research Society (MRS) Fall Meeting, Boston (Nov 2023).
5. **T. Kumar**, D. Feng, S. Yin, P. Lin, M. Mah, M. Fortman, G. Jaffe, C. Wan, C. Fang, R. Warzoha, V. Brar, J. Talghader, M. Kats, *Photothermal Commonpath Interferometry of Silicon Nitride Membranes for Laser Light Sails*, Conference on Lasers and Electro-Optics (CLEO), San Jose (May 2023).
6. **T. Kumar**, A. Sugathan, K.L. Narasimhan, A. Pandey, S. Avasthi, *CuFeSe₂ Quantum-Dot Based Infrared Photodetectors with Functionality in the Ambient* (Poster), IEEE Sensors Conference (online), Nov 2021. [\[DOI\]](#)
7. **T. Kumar**, R. E. Owyung, S. Sonkusale, *High Throughput Cleanroom-Free Fabrication of Thread Based Transistors*, Materials Research Society Fall Meeting (online), Nov 2020.

Talks

1. *Building a Satellite Isn't Rocket Science*, TEDx BITS Pilani, India, Jan 2019 [\[Link\]](#).
2. *Role of Open Source in the Development of a Student-Built Nanosat & its Team Members*, Open Source Cubesat Workshop - European Space Agency, Madrid, Sep 2018 [\[Link\]](#).

Research Experience

1. Characterization of low-loss laser-sail materials Sep 2021 - ongoing
University of Wisconsin-Madison - with Prof. Mikhail Kats
We are designing and conducting experiments to find the optical parameters of very low-loss materials. These materials have applications in laser sails, to be used in the Breakthrough Starshot project.

2. Quantum Dot Photodetectors Aug 2020 - Jul 2021
Indian Institute of Science - with Prof. Sushobhan Avasthi
 We designed the fabrication flow and materials for ambient-operation broadband CuFeSe₂ and CuFeS₂ quantum dot photodetectors.
3. Rapid Fabrication Process for Thread-Based Transistors Aug 2019 - June 2020
Tufts University - with Prof. Sameer Sonkusale
 We developed a high-throughput, cleanroom-free fabrication method for Thread-Based Transistors (TBTs). This improved on the throughput of a previous fabrication method by 2 orders of magnitude.

Internships

1. CentraleSupélec, Paris May 2019 - July 2019
With Prof. Caroline Lelandais-Perrault
2. Indian Space Research Organization May 2018 - July 2018
With Dr. Vinod Kumar, FIE, FIETE, Scientific Officer

Teaching Assistantships (During Undergraduate)

1. Microelectronic Circuits (EEE F244) Jan 2019 - May 2019
 - Designed and conducted tutorial sessions on SPICE & Cadence simulation techniques and graded tests
 - Group coordinator for my TA group.
2. Computer Programming (CS F111), Aug 2018 - Dec 2018
 - Conducted twice-a-week lab classes on Linux commands and C programming for freshman year students.
 - Developed problem sets and answer keys and invigilated quizzes.

Professional Service

1. Referee for Advanced Materials Technologies
2. President, Graduate Student Association (Sep 2023 - Aug 2024)
 Department of Electrical and Computer Engineering (ECE-GSA), University of Wisconsin-Madison
3. Panelist, Graduate student life panel, Visit WI ECE for prospective students
4. Panelist, Graduate student life panel, Graduate School, UW-Madison
5. Volunteer for Ask-a-Grad-Student program for undergraduate students

Music and Photography

Available on my website - <https://tanuj.co>